

3.3.3 The Quotient Rule

Recall: Last class we learned that the *Product Rule* states: $\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$

We will now address the problem of differentiating the quotient of two functions.

Example 1: Use the definition of the derivative to find a formula for $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right]$ assuming that both f and g are differentiable and $g(x) \neq 0$.

We calculate that:

$$\begin{aligned}
 \frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] &= \lim_{h \rightarrow 0} \frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} = \lim_{h \rightarrow 0} \frac{\frac{f(x+h)g(x) - f(x)g(x+h)}{g(x+h)g(x)}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{hg(x+h)g(x)} \\
 &= \lim_{h \rightarrow 0} \frac{g(x)(f(x+h) - f(x)) - f(x)(g(x+h) - g(x))}{hg(x+h)g(x)} \\
 &= \lim_{h \rightarrow 0} \frac{g(x)(f(x+h) - f(x))}{hg(x+h)g(x)} - \lim_{h \rightarrow 0} \frac{f(x)(g(x+h) - g(x))}{hg(x+h)g(x)} \\
 &= \lim_{h \rightarrow 0} \left[\left(\frac{g(x)}{g(x+h)g(x)} \right) \left(\frac{f(x+h) - f(x)}{h} \right) \right] - \lim_{h \rightarrow 0} \left[\left(\frac{f(x)}{g(x+h)g(x)} \right) \left(\frac{g(x+h) - g(x)}{h} \right) \right] \\
 &= \frac{f'(x)}{g(x)} - \frac{f(x)g'(x)}{[g(x)]^2} = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}.
 \end{aligned}$$

The above examples provides proof of the following theorem.

Theorem (The Quotient Rule): If f and g are differentiable, then

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$$

Note: The formula for the derivative produced by the quotient rule is valid for all x such that $g(x) \neq 0$.

Example 2: If $g(x) = \frac{2e^x}{x^2}$, then find $g'(x)$.

Using the *Quotient Rule* we compute:

$$g'(x) = \frac{(x^2)(2e^x) - (2e^x)(2x)}{(x^2)^2} = \frac{2xe^x(x-2)}{x^4} = \frac{2e^x(x-2)}{x^3}$$

Notice that we could also simplify before we start and then use the *Product Rule*:

$g(x) = \frac{2e^x}{x^2} = 2e^x(x^{-2})$ Therefore using the *Product Rule* we compute:

$$g'(x) = 2e^x(-2x^{-3}) + 2e^x(x^{-2}) = \frac{-4e^x}{x^3} + \frac{2e^x}{x^2} = \frac{-4e^x}{x^3} + \frac{2xe^x}{x^3} = \frac{2xe^x - 4e^x}{x^3} = \frac{2e^x(x-2)}{x^3}$$

Example 3: Find the first derivative of $f(x) = \frac{2-x}{x^2+6}$.

Using the *Quotient Rule* we compute:

$$\begin{aligned} f'(x) &= \frac{(x^2+6)\frac{d}{dx}(2-x) - (2-x)\frac{d}{dx}(x^2+6)}{(x^2+6)^2} \\ &= \frac{(x^2+6)(-1) - (2-x)(2x)}{(x^2+6)^2} \\ &= \frac{-x^2-6-4x+2x^2}{(x^2+6)^2} \\ &= \frac{x^2-4x-6}{(x^2+6)^2} \end{aligned}$$

Note: Because of the complexity of the denominator of the fraction this problem cannot yet be done using the Product Rule. In order to use the product rule here we will need learn a new derivative rule known as the Chain Rule.

Example 4: Find an equation of the tangent line to the curve $y = \frac{e^x}{1+x^2}$ when $x = 1$.

Using the *Quotient Rule* we compute:

$$y' = \frac{(1+x^2) \frac{d}{dx}(e^x) - e^x \frac{d}{dx}(1+x^2)}{(1+x^2)^2} = \frac{(1+x^2)e^x - 2xe^x}{(1+x^2)^2} = \frac{e^x(x^2 - 2x + 1)}{(1+x^2)^2} = \frac{e^x(x-1)^2}{(1+x^2)^2} = e^x \left(\frac{x-1}{1+x^2} \right)^2$$

Therefore the slope of the tangent line is: $m = y'(1) = e^1 \left(\frac{1-1}{1+1^2} \right)^2 = 0$. (We have a horizontal tangent line)

We also know when $x=1$ we have $y = \frac{e^1}{1+1^2} = \frac{e}{2}$. Therefore we are making a tangent line at the point $\left(1, \frac{e}{2}\right)$.

The equation of the tangent line is given by:

$$y - \frac{e}{2} = 0(x-1) \Leftrightarrow y = \frac{e}{2}$$

Summary of derivative formulas we know so far:

Assume that f and g are differentiable and c is a constant. We have:

- $\frac{d}{dx}[c] = 0$
- $(f \pm g)' = f' \pm g'$
- $\frac{d}{dx}[x^n] = nx^{n-1}$
- $\frac{d}{dx}[e^x] = e^x$
- $(cf)' = cf'$
- $(fg)' = fg' + gf'$
- $\left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$ when $g \neq 0$